7.4 Mapping Archaeologically-sensitive Areas for Municipalities' Comprehensive Plans

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Introduction

Based on the current Heritage Conservation Act (HCA, 2019), elements of the archaeological heritage can be taken under state protection in Estonia as an archaeological monument and it is also possible to designate an area of archaeological value as a protected archaeological site (see section 6.2). The listing procedure for both is time-consuming and labour intensive, and as a result, a large number of certain archaeological sites have accumulated over the past 20 years, which are not yet officially protected (section 6.2, Fig.5), i.e. their existence may not be known to the landowners, but certainly not to planners in local governments or hobby detectorists. This means that all four threats mentioned in the Valletta Convention (comprehensive planning schemes, natural risks, clandestine or unscientific excavations, and insufficient public awareness) are a daily problem in Estonia concerning the archaeological sites that are known, but not yet under state protection.

In 2015-2017, an administrative reform took place in Estonia, after which most of the 79 remaining local governments had to adopt a new comprehensive plan, which must assess among other things the possible impact of planning on cultural heritage. The National Heritage Board (NHB) prepared a guide for local governments preparing the comprehensive plan on how to deal with cultural heritage, including archaeological heritage. In addition to instructions on monuments, attention was also drawn to the archaeological heritage which has not yet been protected. The principle is that the sooner one is aware of the heritage when planning landscape changes, the greater the chance of making better and more inclusive choices. As the local authorities were not willing to organise the mapping of the archaeological heritage themselves, the NHB took on this work. Since 2020, to the point of writing, possible archaeologically sensitive areas have been mapped in 57 municipalities (Fig.24).

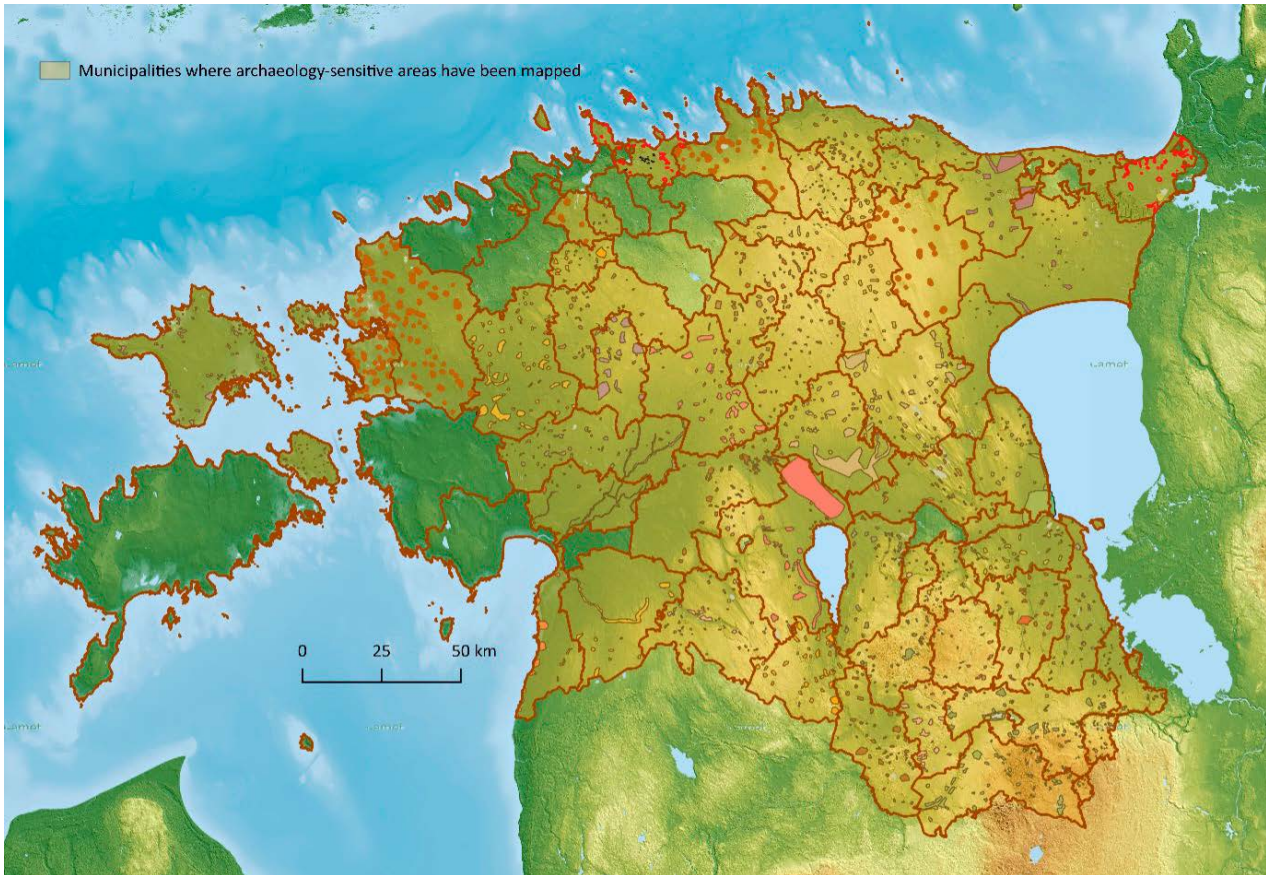

Figure 24: Municipalities where archaeology-sensitive areas have been mapped. (National Heritage Board of Estonia).

Mapping focus

Places with a clear human impact can be everywhere, but their most likely location can be predicted to some extent. Thanks to the archaeological research carried out so far, archaeologists have an idea of where archaeological sites from different periods are likely to be found. For example, the settlements of Stone Age hunters-fishers-gatherers were concentrated mainly on the banks of water bodies, but over the millennia, former beaches are now often deep inland or, in some places, submerged. In Estonia, farming became the predominant way of life during the Bronze Age – households were located sparsely in the landscape, left behind only a weak cultural layer, and are indicated by cup-marked stones and stone-cist graves. The current settlement pattern in Estonia largely developed during the Late Iron Age (6th-13th century AD), i.e. there is a high probability of finding cultural layers of the households of that time in the existing old villages and their surroundings. The burial tradition that has changed over the millennia has left different types of burial sites in the landscape, the location of which in relation to the habitat can also be predicted. In addition to these, there are also sites in the vicinity of habitats that indicate all kinds of economic activity (ancient field systems, iron smelting and processing sites, etc.). For all these sites, it is better to know in advance their existence, since it is impossible to notice fragments of stone flakes and ceramics or human bones, burnt stones, etc from the excavator. The focus of the mapping is on Stone Age hunter-gatherer and Bronze, Iron Age and medieval agricultural settlements and their burial places (Kadakas et al, 2021).

Mapping method

The mapping of archaeologically sensitive areas is based on a compilation of available data and a map-based analysis, which results in the delineation of areas most likely to contain archaeological features.

The mapping of archaeologically sensitive areas has been made possible thanks to a large amount of preparatory work carried out for other purposes within the framework of various research projects. The main source is the geoinformation database of Estonian archaeological sites, compiled by the University of Tartu, which for years has collected all available archaeological records, including information from archaeological surveys and stray finds in museum collections, as well as any occasional mentions of archaeological sites in archives or published sources. The Institute of the Estonian Language has compiled a database of place names, which also includes the date of the first mention of the name (in the Estonian context, the oldest written sources start from the 13th century). The NHB collects information on finds discovered by hobby detectorists and the places mainly visited by licenced detectorists[22]. Historical and current maps are used to analyse the site data. The National Archives of Estonia's public map server has many digitised historical maps from the 17th century onwards and the Estonian Land Board geoportal has current relief maps and aerial photographs from the 1940s-1950s, which provide information on the landscapes before the massive land reclamations during the Soviet occupation.

Purpose and legal framework

The main purpose of the mapping of archaeologically sensitive areas is to inform the public about the archaeological heritage and to create the preconditions for taking this heritage into account in large-scale planning. However, the archaeologically sensitive areas cover quite large territories compared to actual archaeological sites, i.e they are not very precise and include many possible sites as well as "blank" plots in between them. These areas are subject to two somewhat contradictory considerations: 1) their designation must allow to take archaeological sites which are not yet listed into account when planning; 2) their designation must be sufficiently general to prevent looting on these not yet protected sites. The purpose of the mapping is not to give precise estimates of properties, but to indicate broad areas based on archaeological sources, so that during the planning procedure, advice can be sought from the NHB.

[22]In Estonia, after completing the training, hobby detectorists can apply for a permit to search from the NHB. During the training, they will be introduced to the nature of archaeological heritage, the main types of finds and how to proceed when they find something. Detectorists are allowed to pick up finds from the ploughed layer, but they are not allowed to excavate in situ finds, which can only be excavated by an archaeologist. Detectorists must inform the National Heritage Board of the search time and the location, and submit a report afterwards.

The setting of special conditions to archaeologically sensitive sites is currently optional for local authorities, as there is no obligation to do so under the current HCA. However, the Planning Act allows to set them and most municipalities, in cooperation with the NHB, have stipulated in their comprehensive plan that, as part of the environmental impact assessment, as well as when initiating detailed plans or granting permission for the construction of a building or complex larger than 500 m^2, the opinion of the NHB on the need to carry out an archaeological survey in an archaeologically sensitive area is sought. The NHB can then reassess the feasibility of commissioning a preliminary survey or can advise the owner on what could be found during the build and when to inform archaeologists.

Conclusion

The Estonian HCA, essentially referring to the Valletta Convention, provides for two main values of archaeological monuments: they are indicators of the temporal multilayered nature of the cultural landscape (carriers of collective memory) and they provide scientific information on the history of humankind and its relationship with the natural environment. These values are to some extent contradictory: if the monument is intact on the landscape, it plays the role of a time capsule, but we gain information only by exploring, i.e. excavating, but in doing so we are reducing the authenticity of the object in the landscape. Therefore, in assessing archaeological heritage we must constantly consider whether and how much to explore or preserve. However, in decision-making only what is known and what is acknowledged can be considered.